

# Sangwook Lee

Ph.D. Candidate, Department of Computer Science, Virginia Tech (echolab)

sangwooklee@vt.edu · sangwooklee.info · Google Scholar · github.com/woogler · +1-540-250-8969

---

## Research Interests

I design, build, and deploy LLM-based agents that **bridge gaps between people**, spanning human-AI interaction, social computing, and accessibility, while keeping end users in control of what agents do on their behalf:

- **Knowledge gap** — turning everyday lab communication into organizational memory (CHOIR, CHI '26)
- **Digital divide** — adapting GUIs and navigation to each user's preferences (MAESTRO, UIST '26)
- **Expertise gap** — letting domain experts, not engineers, shape AI behavior (ModSandbox, CHI '23; SCORE)

## Education

- 08/2024 – Present     **Virginia Tech**, Blacksburg, VA  
Ph.D. Candidate in Computer Science (**expected graduation: 05/29**)  
Advisor: Sang Won Lee
- 09/2021 – 02/2024     **Korea Advanced Institute of Science and Technology (KAIST)**, Daejeon, Korea  
M.S. in Computer Science. Advisor: Insik Shin  
Thesis: *TOMAS: A Task-Oriented Multimodal Agent System to Enhance Mobile User Interface Accessibility for Older Adults*
- 03/2014 – 08/2020     **Pohang University of Science and Technology (POSTECH)**, Pohang, Korea  
B.S. in Convergence IT Engineering, *Cum Laude*

## Research Experience

- 08/2024 – Present     Graduate Research Assistant, **echolab, Virginia Tech**, Blacksburg, VA  
Advisor: Sang Won Lee. Collaborators: Young-Ho Kim (NAVER AI Lab), Yan Chen (Toronto)  
**CHOIR** — *chatbot-mediated organizational memory from lab communication [P4]*  
LLM Slack chatbot for document-grounded Q&A and AI-assisted document updates; one-month deployment in 4 research labs (N=21). CHI '26 Best Paper Honorable Mention.  
**MAESTRO** — *conversational agent that adapts GUIs to user preferences [P5]*  
Preference memory drives in-place GUI adaptation and workflow backtracking; a within-subjects study (N=33) showed fewer preference violations than a baseline agent; UIST '26 paper and demo.  
**SCORE** — *configuring classroom LLM chatbots without prompt engineering (in progress)*  
Instructors split a course chatbot's instructions by kind of student question (intent → rule) instead of one monolithic prompt, previewed on real student-chatbot logs; a within-subjects study with writing instructors is under way. Built on SWAG, my writing platform (101 students).  
**RYLAI** — *chatbot role-play training against cybergrooming for adolescents (NSF SaTC #2330940)*  
Sole developer of the web platform and the self-hosted model server running the team's StagePilot policy (An et al., SIGDIAL '26) on a Qwen2.5-7B generator; deployed under IRB.
- 09/2021 – 02/2024     Graduate Research Assistant, **Cyber-Physical Systems Lab, KAIST**, Daejeon, Korea  
Advisor: Insik Shin  
*Mobile UI automation and accessibility: multi-app UI mashup [P1]; M.S. thesis on TOMAS, a multimodal agent that simplifies smartphone UIs for older adults [T1].*
- 06/2023 – 08/2023     Visiting Student, **SaNDwich Lab, University of Notre Dame**, Notre Dame, IN  
Mentor: Toby Jia-Jun Li  
*Led the design and implementation of TOMAS [T1] with the OpenAI API; mentored two undergraduate interns.*

- 09/2020 – 09/2021      Research Intern, **KIXLAB, KAIST**, Daejeon, Korea  
Mentors: Juho Kim, Jean Y. Song  
*Led ModSandbox, a sandbox that predicts errors of rule-based moderation before deployment; formative interviews and a between-subjects study with 20 Reddit moderators [P2].*
- 07/2017 – 08/2017      Research Intern, **HCI+D Lab, Seoul National University**, Seoul, Korea  
Mentor: Joonhwan Lee  
*Participatory design workshop with female gamers on harassment in online games [W3].*

## Industry Experience

- 02/2024 – 08/2024      Software Engineer, **Fluiz, Inc.**, Seoul, Korea  
KAIST faculty startup building mobile AI agents; Samsung Electronics C-Lab Outside (2026)  
*Designed and prototyped LLM-based workflow automation for judicial scriveners with legal-software partner NSSAM: a Chrome extension records an expert's interactions and rationale, an LLM compiles them into reusable document-issuance workflows, and a remote Playwright server executes them.*

## Publications

### Peer-Reviewed Conference and Journal Papers

- [P5]      **Sangwook Lee**, Adnan Abbas, Sang Won Lee, Young-Ho Kim, and Yan Chen. 2026. [MAESTRO: Adapting GUIs and Guiding Navigation with User Preferences in Conversational Agents with GUIs](#). *Proceedings of the ACM Symposium on User Interface Software and Technology (UIST '26)*, to appear Nov. 2026.
- [P4]      **Sangwook Lee**, Adnan Abbas, Yan Chen, Young-Ho Kim, and Sang Won Lee. 2026. [CHOIR: A Chatbot-mediated Organizational Memory Leveraging Communication in University Research Labs](#). *Proceedings of the ACM Conference on Human Factors in Computing Systems (CHI '26)*.  
★ **Best Paper Honorable Mention (Top 5%)**
- [P3]      Donghan Hu, Junghoon Chung, Daniel Vargas Diaz, **Sangwook Lee**, Sang Won Lee, and Sol Lim. 2026. [Toward Context-Aware and Personalized Sit-Stand Desk Interventions: Insights from a Field Observational Study](#). *International Journal of Human-Computer Studies*, 215.
- [P2]      **Sangwook Lee\***, Jean Y. Song\*, Jisoo Lee, Mina Kim, and Juho Kim (\*equal contribution). 2023. [ModSandbox: Facilitating Online Community Moderation Through Error Prediction and Improvement of Automated Rules](#). *Proceedings of the ACM Conference on Human Factors in Computing Systems (CHI '23)*.
- [P1]      Sunjae Lee, Hoyoung Kim, Sijung Kim, **Sangwook Lee**, Hyosu Kim, Jean Y. Song, Steven Y. Ko, Sangeun Oh, and Insik Shin. 2022. [A-Mash: Providing Single-App Illusion for Multi-App Use Through User-centric UI Mashup](#). *Proceedings of the ACM International Conference on Mobile Computing and Networking (MobiCom '22)*.

### Thesis

- [T1]      **Sangwook Lee**. 2023. [TOMAS: A Task-Oriented Multimodal Agent System to Enhance Mobile User Interface Accessibility for Older Adults](#). *M.S. thesis, KAIST (Advisor: Insik Shin), with Qianhui Zhao, Haoming Ma, and Toby Jia-Jun Li*.

### Workshop Papers, Posters, and Extended Abstracts

- [W6]      **Sangwook Lee** and Sang Won Lee. 2026. [Redistributing Interdependence in Organizational Knowledge Work: Lessons from Deploying an LLM Mediator in Research Labs](#). *Co-Data: Cultivating Effective Human-LLM Collaboration for Collaborative Data Processing, Workshop at CHI '26*.
- [W5]      **Sangwook Lee** and Sang Won Lee. 2026. [TUNING: Adaptive GUI Modification as Agent Tools in Task-Oriented Chatbots](#). *Human-AI-UI Interactions Across Modalities, Workshop at CHI '26*.
- [W4]      Hansoo Lee, **Sangwook Lee**, Youngji Koh, and Uichin Lee. 2022. [LV-Linker: Supporting Fine-grained User Interaction Analyses by Linking Smartphone Log and Recorded Video Data](#). *Adjunct Proceedings of the ACM Symposium on User Interface Software and Technology (UIST '22 Adjunct)*.

- [W3] Wookjae Maeng, Hyeok Kim, Junhee Woo, Youngjin Huh, Seo-young Lee, Jeewon Choi, **Sangwook Lee**, Jinsu Eun, Kung Jin Lee, and Joonhwan Lee. 2018. [Implications for the Design of Sexual Harassment Prevention System for Online Games](#). *Proceedings of HCI Korea 2018*.
- [W2] **Sangwook Lee** and Hyunggu Jung. 2016. [Mutiv: Music-based Mobile Application to Support Joggers](#). *Poster, IEEE International Conference on Healthcare Informatics (ICHI '16)*.
- [W1] **Sangwook Lee**, Yunho Kang, and Yukyoung Lee. 2016. [LaneMate: Car Sensing System for the Deaf](#). *Extended Abstracts of the ACM Conference on Human Factors in Computing Systems (CHI EA '16)*.  
**★ Student Design Competition Honorable Mention (Final Round)**

## Awards & Honors

|      |                                                                                       |
|------|---------------------------------------------------------------------------------------|
| 2026 | <b>Best Paper Honorable Mention</b> (Top 5%), ACM CHI 2026 [P4]                       |
| 2026 | <b>Special Recognition for Outstanding Reviews</b> (×3), ACM CHI 2026                 |
| 2022 | <b>Gary Marsden Travel Award</b> , ACM SIGCHI                                         |
| 2016 | <b>Honorable Mention</b> , ACM CHI 2016 Student Design Competition (Final Round) [W1] |

## Academic Service

|             |                                                                                       |
|-------------|---------------------------------------------------------------------------------------|
| 2023 – 2026 | <b>Reviewer</b> — ACM CHI 2023–26 (3 Outstanding Reviews, 2026), UIST 2026, CSCW 2024 |
| 2020 – 2026 | <b>Student Volunteer</b> — CHI 2021–26 (Photo SV 2024–26, Video Lead 2025), CSCW 2020 |

## Mentoring

|                |                                                                                       |
|----------------|---------------------------------------------------------------------------------------|
| 2026 – Present | Joshua Hairston, M.S. student, Virginia Tech (SCORE)                                  |
| 2023           | Chloe Qianhui Zhao, undergraduate intern, Univ. of Notre Dame (now Ph.D. student, UW) |
| 2023           | Haoming Ma, undergraduate intern, University of Notre Dame (now ML Engineer, TikTok)  |

## Talks

|         |                                                                                                                                                                                  |
|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 05/2026 | <i>Bridging Gaps Between People with AI-Based Agents</i> (thesis seminar). <b>HCI+D Lab, Seoul National University</b> (Joonhwan Lee); <b>HAI Lab, SeoulTech</b> (Kyoungwon Seo) |
| 03/2026 | Opener talk, <b>CHCI@VT Weekly Seminar</b> , Virginia Tech                                                                                                                       |
| 07/2023 | <i>Leveraging NLP Models in Practice</i> . <b>DGP Lab, University of Toronto</b> (SuHyeon Yoo)                                                                                   |

## Teaching

|                    |                                                                                                                                                                            |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Teaching Assistant | CS 4104 Data and Algorithm Analytics, Virginia Tech (Fall 2025); CS 584 Human-Computer Interaction, KAIST (Fall 2023); CS 330 Operating Systems and Lab, KAIST (Fall 2021) |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## Skills

|             |                                                                                  |
|-------------|----------------------------------------------------------------------------------|
| Methods     | Experiments, field deployments, interviews, surveys; GLMM/LMM, thematic analysis |
| Engineering | Python, TypeScript, R; React/Next.js, Django; OpenAI/Anthropic APIs, RAG, Ollama |